

Serin Baek

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EDUCATION

University College London: MSci Pharmacology
MSci Pharmacology (First Class Honors, on track)

2023 - present

RESEARCH POSITIONS

Fitzgerald's Group, Early Cancer Institute University of Cambridge

July 2026 - August 2026

Supervised by Professor Rebecca Fitzgerald and Dr. Ahsen Ustaoglu

Conducted translational research evaluating non-endoscopic capsule sponge assays and liquid biopsies for the early detection of oesophageal adenocarcinoma. Performed ultra-sensitive protein quantitation (SIMOA, ELISA) and bioinformatic methylation analyses (EM-Seq vs. 5-Base) to identify and validate candidate protein and epigenetic biomarkers.

Moss Group, UCL

June 2026 - September 2026

Supervised by Dr. Guy Moss and Dr. Thomas Blacker

Conducted patch clamping and fluorescence imaging by transfecting HEK293 cells with JEDI2P and KCNK2 channels, in order to investigate the bioelectric profile of airway epithelia to innovate gene therapy for cystic fibrosis.

Sihra, UCL

September 2025 - June 2026

Supervised by Professor Talvinder Sihra

Reviewed 82 peer reviewed studies in a qualitative synthesis, writing a 5000 word review on Phosphorylation Dependent Modulation of Presynaptic Plasticity.

Centre for Cardiovascular and Metabolic Neuroscience (CCMN), UCL

January 2025 - April 2025

Supervised by Professor Alexander Gourine and Dr. Alice Braga

Explored the role of adenosine signalling in astrocytic regulation of metabolism. Performed cryostat sectioning and staining on mice brains.

WORK EXPERIENCE

Operational Excellence Summer Intern, AstraZeneca UK

June 2025 - September 2025

Designed a Power BI dashboard from Businessmap to track KPIs, developed Excel VBA automation for SAP processes, and analysed 3,000+ Supplier Quality Risk Assessments to identify risk trends, supporting compliance, patient safety, and senior-level decision making. Received a Gold certificate on Thriving in the age of AI.

LEADERSHIP EXPERIENCE

Student Ambassador, UK Council for International Student Affairs (UKCISA)

December 2025 - present

Student ambassador of UK's national advisory board, providing lived-experience insight to improve support for international students in the UK. Attended global education and policy forums as a guest speaker, including The PIE Live and UKCISA Annual Conference.

Room Monitor Supervisor, UCL

October 2024 - present

Oversee all room monitors of UCL's administrative office to ensure operational readiness. Accumulated 300+ hours of early-morning room monitoring (8AM start), demonstrating consistency and diligence.

Student President, East-West International School, Cambodia

August 2021 - Dec 2022

Initiated a volunteering partnership with Cambodia Children's Fund, designed 200+ student report card templates and developed Excel-based class timetables in collaboration with fellow interns.

VOLUNTEERING EXPERIENCE

Homeless Outreach, Euston Church

October 2025 – May 2026

Baked desserts and spoke to homeless people every other week, acting as the listening ears while gaining insight into how social vulnerability increases health challenges.

Volunteering and Welfare Officer, Music Society, UCL

September 2024 - June 2025

Organised 4+ volunteering concerts at hospitals and schools, coordinating logistics and performance schedules with UCL Student Union and musicians.

Mobile Clinic Medical Volunteer, One Peninsula Medical Union, Cambodia

July 2024

Supported clinicians in safe and effective dosing decisions in a resource-limited setting, interpreting (Khmer) between doctors and patients and ensuring clear communication.

AWARDS AND FUNDING

Dean's List, University College London

August, 2026

Awarded in recognition of exceptional academic performance in the Faculty of Life Sciences (top 5%).

Early Cancer Institute Summer Programme sponsored by Moderna

July, 2026

Fully funded an 8-week summer studentship at the Early Cancer Institute at the University of Cambridge.

Cystic Fibrosis Trust Summer Studentship Scheme, Cystic Fibrosis Trust UK

July, 2026

Conducted research on bioelectric profiling of airway epithelial cell types in cystic fibrosis.

D.H. Jenkinson Prize, University College London

September, 2024

Awarded to the top-performing student in the MSci/BSc Pharmacology cohort for best overall first-year academic performance (81% average).

Presidential Science Scholar (Top 20 Nationwide), South Korea

2023-2027

Awarded the most prestigious scholarship by the South Korean government, fully funding undergraduate studies abroad (approx. \$200,000 USD), with no bond requirement.

TEACHING EXPERIENCE

East-West International School

September 2022 – December 2022

Assistant teacher for A-Level Further Mathematics and IGCSE Mathematics, having graduated as valedictorian with an A* in A-Level Maths.

RESEARCH TALKS

SKILLS

Programming: R, Python

Electrophysiology & Wet Lab: Western Blot, molecular cloning, pharmacological dose-response assays, patch clamp recording, mammalian cell culture, neuronal cell culture, transfection, fluorescence microscopy, SIMOA, ELISA, BCA Protein Assay

Imaging & Analysis: Andor Solis, Fiji (ImageJ), GraphPad Prism, HEKA Pulse, Excel